FEUP-M.EIC001-2026/2027-1S

Marcar como concluída

Abre: sexta-feira, 27 de novembro de 2026 às 09:00
Data limite: sexta-feira, 4 de dezembro de 2026 às 23:59

Introduction

This project consists of the development of a data mining (DM) case study. The description of the application is provided in a separate page that can be found in moodle.

The project consists of a predictive data mining task, which is defined in the case study description. The development of the project should be guided by a suitable methodology. The choice of methodology is left to the students, based on what was presented in class.

Rules

- The project will be developed by groups of 3 students.
- There will be a single delivery, to be made according to the instructions below and on the date indicated in the plan. Additionally, the plan contains dates when other results are expected, in order to help students monitor the progress of their work.

Grading

The grade is given by the following formula: (75% SA + 25% P) * IF

- SA=submitted assignment
- P=presentation
- IF=individual factor

Individual factor

The individual factor should always be equal to 1, for every student. This reflects the contribution of each student to the group project. We understand, however, that under certain (rare) circumstances, not everyone deserves a score of 1. In those scenarios, values different from 1 could be acceptable. The value proposed by the group for each element will be considered, but the final decision concerning the individual grade is up to the lecturer(s). The values should be included in the cover page of the report. If the groups cannot reach a consensus, then you should contact the lecturer(s).

Monitoring the progress of your work

As the groups interact with the lecturer, the corresponding grid will be updated and a preliminary grade will be available. The purpose of this grade is for the groups to monitor their progress and the effort that is still necessary to invest, as well as to understand the strengths and limitations of the current work. However, the final grid and, thus, the final grade, will only be available after the final submission of the project.

Presentation/report guidelines

The presentation will serve also as the report. On one hand his means that it must be prepared for a 15 min. talk so it must focus on the main issues of the project. On the other hand, it should also summarize all the work that was carried out, which means that additional information that cannot be presented orally can be included as annexes to the presentation.

Suggested organization:

- cover page, including IF of all the students in the group

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- domain description (1 short slide, as everyone is working on the same domain)
- exploratory data analysis: very short summary, focusing on the main findings (2 slides)
- for the predictive data mining problem
    - problem definition (1 slide)
    - data preparation: very short summary, focusing on the main operations (2 slide)
    - experimental setup (1 slide)
    - results (2 slide)
- conclusions, limitations and future work (1 slide)
- annexes
- TOTAL: max 40 slides

Submission instructions

- The presentation as well as any files that are required to reproduce the results (data, code, etc), will be submitted in moodle.

Estado do trabalho

Grupo	G51
Número da tentativa	Tentativa 1
Estado da submissão	Nenhuma submissão efetuada
Estado da avaliação	Sem avaliação
Tempo restante	Tempo restante: 43 dias 10 horas
Última modificação	-
Comentários à submissão	Comentários (0)

Critérios de avaliação

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BU: analysis of requirements with the end user	not done	minimal attempts to interact	-	limited to determining the business objectives	-	extensive identification of business objetives and additional relevant business information	beyond expectations
BU: definition of business goals	not done	essentially incorrect/ very incomplete business goals	-	essential goals were identified	most goals identified after many trial and error iterations	clearly and quickly identified goals	beyond expectations
BU: translation of business goals into data mining goals	not done	essentially incorrect/ very incomplete translation	not aligned	essential DM goals aligned with business goals	DM goals aligned with business goals after many trial and error interactions	DM goals clearly aligned with business goals	beyond expectations
DU: diversity of statistical methods	not done	single or very limited set of methods	-	diverse but unjustified set of methods	-	rich and justified set of methods	beyond expectations
DU: complexity of statistical methods	not done	simple 1D and 2D methods	complex 3+D methods with	-	complex 3+D methods with	integrated, complex 3+D	beyond expectations

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			unclear results		clear results	methods with clear results	
DU: interpretation of results of statistical methods	not done	incorrect	-	sometimes correct but often incorrect	-	generally correct	beyond expectations
DU: knowledge extraction from results of statistical methods	not done	trivial knowledge obtained	-	interesting OR novel and non-trivial knowledge obtained	-	interesting, novel and non-trivial knowledge obtained	beyond expectations
DU: diversity of plots	not done	single or very limited set of methods	-	diverse but unjustified set of methods	-	rich and justified set of methods	beyond expectations
DU: complexity of plots	not done	simple 1D and 2D methods	complex 3+D methods with unclear results	-	complex 3+D methods with clear results	integrated, complex 3+D methods with clear results	beyond expectations
DU: presentation	not done	confusing, unstructured set of visualizations	-	simple, clear information, even if not entirely well-structured	-	well-structured information	beyond expectations
DU: interpretation of plots	not done	incorrect	-	sometimes corre	-	generally corre	beyond expe

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				ct but often incorrect		ct	ctations
DU: visual knowledge extraction	not done	trivial knowledge obtained	-	interesting OR novel and non-trivial knowledge obtained	-	interesting, novel and non-trivial knowledge obtained	beyond expectations
DP: data integration	not done	-	-	-	conversion of different formats of data values OR matching entities between different sources	conversion of different formats of data values AND matching entities between different sources	beyond expectations
DP: assessment of dimensions of data quality	not done	-	2 dimensions (usually accuracy and completeness)	3 dimensions (... consistency or uniqueness)	4 dimensions (... consistency and uniqueness)	6 dimensions	beyond expectations
DP (cleaning): redundancy	not done	-	-	-	removal of some redundant attributes	systematic removal of redundant attributes	beyond expectations
DP (cleaning): missing data	not done	remove a signif	replace MVs	replace MVs	replace MVs	replace MVs	beyond exp

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		icant number of columns and/or rows	with constant	with statistically simple method (e.g. mean, mode)	with complex method (e.g. regression, classification) with incorrect experimental setup	with complex method (e.g. regression, classification) with correct experimental setup	ctations
DP (cleaning): outliers	not done	-	identify outliers	identify and discuss outliers	identify and discuss outliers, and address them with simple approach	identify and discuss outliers, and address them with complex (technical or domain-dependent) approaches	beyond expectations
DP: data transformation for compatibility with algorithms	not done	-	inadequate, simple discretization OR rescaling	inadequate complex or adequate simple discretization OR rescaling	-	adequate complex discretization or rescaling	beyond expectations
DP: feature engineering from tabular data	not done	-	simple	-	complex	complex	beyond

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			methods and knowledge (e.g. ratios)		methods (e.g. aggregation) OR knowledge (e.g. business concepts)	methods (e.g. aggregation) AND knowledge (e.g. business concepts)	expectations
DP: sampling for domain-specific purposes	not done	inadequate use of the whole sample	focus on inadequate subset of population	-	-	focus on adequate subset of the population	beyond expectations
DP: sampling for development	not done	-	use insufficient sample	non-systematic use of sampling	-	start with very small sample and grow to significant one	beyond expectations
DP: imbalanced data	not done	-	use basic resampling methods incorrectly (e.g. whole dataset)	use basic resampling methods correctly	used advanced methods (e.g. SMOTE) incorrectly (e.g. whole dataset)	used advanced methods (e.g. SMOTE) correctly	beyond expectations
DP: feature selection	not done	use filter-based methods incorrectly (e.g. whole	use filter-based methods correctly	use wrapper-based methods incorrectly (e.g. whol	use wrapper-based methods correctly	correct, combined use of filter and wrapper-based	beyond expectations

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		datas et)		whol e datas et)		meth ods	
predictive: diversity of tasks	not done	-	-	-	classi ficati on OR regre ssion	classi ficati on AND regre ssion	beyo nd expe ctati ons
predictive: diversity of algorithms	not done	teste d 1 algori thm	-	teste d 3 algori thm	-	teste d 4+ with signif icantl y differ ent langu age bias OR with a signif icant num ber of varia nts	beyo nd expe ctati ons
predictive: parameter tuning	not done	-	-	limite d, manu al appr oach	-	syste matic appr oach	beyo nd expe ctati ons
predictive: understanding algorithm behavior	not done	signif icant confu sion about the beha vior of all agori thms	basic unde rstan ding of <3 algori thms	solid (even if not deep) unde rstan ding of <3 algori thms, witho ut unde rstan ding the effec t of the para mete rs	-	solid (even if not deep) unde rstan ding of the beha vior of the major ity of the algori thms used OR of <3 algori thms, also unde	beyo nd expe ctati ons

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						rstanding the effect of the parameters	
predictive (performance estimation) training vs test	not done	training and testing on the same data	training and testing on separate data but with leakage	-	training and testing on correctly separated data	training and testing on correctly separated data, with multiple splits	beyond expectations
preditive (perfomance estimation): other factors (e.g. time)	not done	additional factors not taken into account	-	additional factors taken into account but not correcly	-	additional factors correctly taken into account	beyond expectations
predictive (performance estimation): perfomance measure	not done	focus on performance measure(s) neither aligned with DM goals nor data characteristics	-	focus on performance measure(s) aligned with DM goals but not data characteristics	-	focus on performance measure(s) aligned with DM goals and data characteristics	beyond expectations
predictive (performance estimation): correct interpretation of performance measures	not done	-	-	basic performance measures (e.g.	medium complexity performance	advanced performance measures	beyond expectations

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				accuracy, error, MSE, MAD)	measures (e.g. precision, recall, ...)	(e.g. AUC)	
predictive (performance estimation): baseline	not done	-	-	unsuitable baseline	-	suitable baseline	beyond expectations
predictive (performance estimation): analysis of results	not done	-	incorrect analysis of values for comparison	-	correct analysis of values for comparison, including informal discussion of statistical significance	correct analysis of values for comparison, including tests for statistical significance	beyond expectations
predictive (performance estimation): analysis of overfitting	not done	incorrect estimation of overfitting	-	correct estimation of overfitting, analyzed incorrectly	-	correct estimation of overfitting, correctly analyzed	beyond expectations
predictive: model improvement	not done	1 iteration, involving a limited set of experiments	-	random set of experiments, without concern for pedagogical goals	random set of experiments, with total focus on pedagogical goals	development guided by performance improvement goals, even if peda	beyond expectations

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						gogical goals were not ignored)	
predictive: feature importance	not done	-	interpreted incorrectly, without relating to application domain	interpreted incorrectly, relating to application domain	interpreted correctly , without relating to application domain	interpreted correctly, relating to application domain	beyond expectations
predictive: analysis of "white-box" models	not done	-	interpreted incorrectly, without relating to application domain	interpreted incorrectly, relating to application domain	interpreted correctly , without relating to application domain	interpreted correctly, relating to application domain	beyond expectations
project management: methodology	not done	-	a methodology was mentioned but the development of the project didn't seem to follow it	the main steps of a methodology were followed loosely	-	a methodology was followed carefully, reflecting on the documentation produced	beyond expectations
project management: plan	not done	-	a plan was created	a rough plan was	-	a detailed plan was	beyond expectations

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			but there was no concern to follow it	created and it guided the development		created and it guided the development	
project management: PM tools	not done	simple,coarse task sharing	-	detailed task sharing, not supported by appropriate management tool	-	detailed task sharing, supported by appropriate management tool	beyond expectations
project management tools: collaboration tools	not done	complex, unreliable data and workflow sharing	easy communication and unreliable data and workflow sharing	reasonable collaboration with simple collaboration tools	easy communication and documentation, and reasonable collaboration with simple tools	easy collaboration (communication, data, workflows and documentation)	beyond expectations
tools: analytics	not done	-	1 tool, from the ones presented in the courses	-	-	multiple tools, from the ones presented in the courses	beyond expectations
tools: database	not done	only as master	-	used for simple data preparation	-	extensive use for efficient data explo	beyond expectations

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				operations		ration and preparation	
tools: other tools (e.g. data cleaning, visualization)	not done	-	unsuitable use of other tools	-	limited but correct use of other tools	extensive and correct use of other tools	beyond expectations
presentation: quality of layout	not done	hard to understand	-	simple OR clear	-	simple AND clear	beyond expectations
presentation: quality of content in slides	not done	extremely incomplete or excessive	-	incomplete or too much content	-	to the point	beyond expectations
		ation					

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